

ChartPRM: Process Reward Modeling and Step-Level Preference Optimization for Complex Multimodal Chart Reasoning

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Abstract

Can a compact 3B vision-language model reason over complex scientific charts, not just read numbers off them? We study Qwen2.5-VL-3B-Instruct on two benchmarks: 500 reasoning problems from CharXiv and a matched pool from ChartQA. A multimodal LLM-as-a-judge (muse-spark-1.1) scores each reasoning step rather than only the final answer. Over 80% of first errors happen within two steps, and once a step fails, the next fails 83% of the time. Critique analysis of 2,920 failed steps points to perception, not arithmetic: axis and legend misreadings dominate. We train seven alignment methods on this signal (SFT, full DPO, Step-DPO, KTO, SimPO, Pareto-DPO, and SFT→DPO) against the base model and ChartGemma. To keep this supervision from resting on one judge’s scalar opinion, we cluster its critiques into a reward tree and select preference pairs by Pareto dominance across its criteria. That method wins on both benchmarks: Pareto-DPO reaches 30% exact-match on CharXiv, ahead of a 26% base, and 60% on ChartQA after transfer, against a 43% base, the largest gain of any method on either dataset. Chart reasoning in small models fails on perception rather than logic, and dominance-filtered process reward is what survives the move to a second dataset.

1 Introduction

Reasoning over scientific charts requires multimodal models to interleave visual perception (resolving dual axes, tick scales, and legend mappings) with symbolic deduction (computing rates and comparing extrema). Standard Outcome Reward Models (ORMs) provide sparse feedback only on the final answer (Ouyang et al., 2022), failing to isolate whether an error stems from an early visual misreading or downstream arithmetic. In contrast, Process Reward Models (PRMs) provide dense, step-by-step verification to enable granular credit assignment (Lightman et al., 2023).

In this paper, we investigate a central hypothesis: *Can a compact vision-language model reason effectively over complex scientific charts when aligned with only a small, process-supervised dataset?* Rather than relying on tens of thousands of annotations, we study whether fewer than 150 verified contrastive preference pairs suffice to align a compact model under consumer-grade compute constraints. A second question follows directly from the first: once such process supervision is built for one chart-QA benchmark, does it stay specific to that benchmark, or does it transfer, unmodified, to a second one?

We evaluate this hypothesis on the **CharXiv** benchmark (Wang et al., 2024), and later test transfer on **ChartQA** (Masry et al., 2022). Discarding surface-level descriptive queries, we isolate a stratified subset of **500 multi-step reasoning questions** from CharXiv spanning 8 academic domains and 62 chart types. Operating within a 16GB GPU budget (Nvidia Tesla P100/T4), we benchmark Qwen2.5-VL-3B-Instruct (Bai et al., 2025) using native 16-bit precision (FP16) and LoRA (Hu et al., 2022).

Using an LLM-as-a-judge (muse-spark-1.1) (Meta Superintelligence Labs, 2026; Zheng et al., 2023) with rollout-level batching, we evaluate 1,274 rollouts (4,947 steps). Analyzing 2,920 refuted steps reveals two primary diagnostic findings: (1) **Perceptual Bottleneck:** Failures are overwhelmingly visual: axis/layout misreads (24.0%) and series/color confusion (19.5%) account for 43.5% of failed steps, whereas pure arithmetic errors represent only 1.3% (1 in 77 errors); (2) **Catastrophic Cascades:** 79.7% of flawed traces break down in Steps 0–1, and once a step fails, the subsequent step fails 82.7% of the time ($P(S_{N+1} = 0 \mid S_N = 0) = 82.7\%$).

Leveraging these diagnostics, we benchmark seven alignment strategies on a 100-question held-out set: SFT (70 traces), Full DPO (134 pairs)

(Rafailov et al., 2023), Suffix Step-DPO (54 pairs) (Lai et al., 2024), KTO (84 pos / 252 neg) (Ethayarajh et al., 2024), SFT→DPO, the reference-free SimPO (Meng et al., 2024), and a Pareto-dominance-filtered variant of DPO introduced in this work. Confirming our small-data hypothesis, full-trajectory preference optimization is consistently the strongest paradigm: Full DPO reaches 29.0% accuracy on just 134 pairs (+3.0 pp over the 26.0% base), and Pareto-filtering the same pairs edges this further to 30.0%, the best result on CharXiv. SFT enforces 100% formatting compliance but lowers accuracy to 23.0% via template rigidity, while KTO attains 66.0% latent ground-truth recall in text but suffers structural format collapse.

A single binary judge score, however, is a coarse training signal: it cannot tell a rollout that misreads an axis apart from one that hallucinates a legend entry, even though both receive the same failing label. Motivated by Yin et al.’s Dynamic and Generalizable Process Reward Modeling (DG-PRM) (Yin et al., 2024), we adapt this idea to chart reasoning: we distill the judge’s failure critiques into a reward tree of 33 scored criteria grouped under the 9 error categories above, and instead of collapsing a rollout’s scores to one scalar, we keep a 9-dimensional vector and accept a preference pair only when the correct rollout Pareto-dominates the incorrect one across every category at once. This is what produces the 30.0% result. To test whether this dynamic judge is also generalizable, as DG-PRM’s own name claims, we transfer the reward tree unmodified to ChartQA and retrain all seven methods there: the tree fires on ChartQA content at least as densely as on CharXiv, and Pareto-DPO wins by a wide margin, 60.0% exact-match against a 43.3% base, the largest gain of any method on either dataset. We additionally show the same judge is useful at inference time, without any training: reranking existing rollouts by step-level process reward reaches 27.5% accuracy, ahead of majority-vote self-consistency (21.0%) and random selection (18.4%).

Our main contributions are: (1) We validate the small-data alignment hypothesis on Qwen2.5-VL-3B-Instruct under a single-T4 GPU budget using 134 PRM preference pairs; (2) We uncover an empirical perceptual bottleneck (43.5% visual grounding vs. 1.3% arithmetic) and catastrophic cascades (82.7% failure continuation) across 2,920 refuted steps; (3) We benchmark seven

alignment paradigms and chart the Pareto frontier between format compliance and latent reasoning recall; (4) We adapt DG-PRM to chart reasoning, using Pareto-dominance filtering across 33 criteria to achieve 30.0% exact-match on CharXiv; (5) We demonstrate zero-retuning transfer to ChartQA, where Pareto-DPO achieves 60.0% (+16.7 pp over base); and (6) We show that test-time PRM candidate reranking reaches 27.5% EM, surpassing majority-vote self-consistency.

2 The ChartPRM Framework

The central goal of **ChartPRM** is to verify, diagnose, and align step-by-step visual reasoning on complex scientific charts. Scientific plots pack dense information into visual elements. Solving a reasoning question requires multiple distinct steps: locating the relevant panel, reading numerical axis scales, matching curve styles to legend labels, and performing logical comparisons.

ChartPRM decomposes this task into discrete reasoning steps, scores each step individually with a multimodal judge, and uses those step-level signals to guide preference optimization. Crucially, we also ask whether it is possible to align a compact 3B vision-language model using only a very small dataset of verified chart reasoning traces (fewer than 150 examples).

Below, we detail our reasoning protocol, rollout-batched multimodal judge, preference curation, alignment objectives, DynamicPRM tree construction, and test-time verification.

2.1 Structured Step-by-Step Reasoning Protocol

To enable step-level verification, the vision-language model generates Chain-of-Thought reasoning traces (Wei et al., 2022) structured into sequential steps (Step 1:, Step 2:, ...) ending with a concise prediction on its own line (Final Answer: <answer>) without conversational preambles (see Appendix B for the exact prompt template). Candidate rollouts are sampled at temperature $\tau = 0.7$ and top- $p = 0.9$ (up to four rollouts per question), yielding 1,274 complete rollouts across 500 CharXiv (Wang et al., 2024) reasoning problems. An automated parser extracts the prediction following Final Answer: and applies lowercase whitespace normalization to assess exact match against the ground truth.

2.2 Multimodal Process Reward Modeling

We evaluate intermediate reasoning steps using `muse-spark-1.1` (Meta Superintelligence Labs, 2026; Zheng et al., 2023) as a multimodal LLM judge. To overcome token and upload limits, we introduce *rollout-level batching*: we resize the chart image I to ≤ 512 px (retaining visual legibility while minimizing token usage) and evaluate all steps $(y_1, \dots, y_T)$ within a single request alongside the question q and reference answer y^* .

For each step t , the judge returns: (1) a **Binary Step Score** $s_t \in \{0, 1\}$, where $s_t = 1$ indicates correct visual grounding and deduction, while $s_t = 0$ penalizes axis misreads, color confusion, hallucinated elements, or calculation errors; and (2) a **Natural Language Critique** detailing the failure mode for failed steps. The overall process reward of trajectory τ is the mean step score:

$$R(\tau) = \frac{1}{T} \sum_{t=1}^T s_t \quad (1)$$

A trajectory is *fully verified* only if every step passes ($R(\tau) = 1.0$).

2.3 Preference Dataset Curation and Step Slicing

From 1,274 judged rollouts (4,947 steps), we curate datasets for distinct alignment paradigms:

1. SFT Demonstrations: We select rollouts with perfect process scores ($R(\tau) = 1.0$) and correct final answers, yielding 70 gold-standard reasoning traces of verified visual deduction.

2. Sequence DPO Pairs: For questions with multiple rollouts, we form pairs (y_w, y_l) where $R(y_w) > R(y_l)$ or y_w attains the correct final answer when y_l does not, yielding 134 chosen-rejected pairs.

3. Suffix Step-DPO Slicing: Standard sequence DPO penalizes entire trajectories equally, ignoring early correct steps. To isolate credit assignment, Step-DPO identifies the first divergence point $t^* = \min\{t \mid y_{w,t} \neq y_{l,t} \text{ or } s_t(y_l) = 0\}$. Keeping shared prefix $y_{<t^*}$ as unpenalized prompt context, we optimize preference loss strictly over the divergent suffix, yielding 54 prefix-aligned pairs.

4. KTO Unpaired Dataset: Kahneman-Tversky Optimization uses unpaired completions. We partition traces into 84 desirable ($R \geq 0.75$ and correct answer) and 252 undesirable ($R < 0.5$ or wrong answer) rollouts (1:3 positive-to-negative ratio).

Method	Data Size	Epochs	LR	Time
SFT	70 traces	1	1×10^{-5}	12 min
Full DPO	134 pairs	1	1×10^{-5}	22 min
Step-DPO	54 pairs	2	1×10^{-5}	15 min
KTO	84 pos / 252 neg	2	2×10^{-6}	35 min
SFT→DPO	134 pairs	1	2×10^{-6}	24 min
SimPO	134 pairs	1	1×10^{-6}	–
Pareto-DPO	154 pairs	1	1×10^{-5}	–

Table 1: **Fine-Tuning Configurations under 16GB GPU Budget.** All models use FP16 LoRA ($r = 16$) on Qwen2.5-VL-3B-Instruct (Bai et al., 2025).

2.4 Preference Alignment Objectives and Training Setup

We align policy π_θ against unaligned base instruct model π_{ref} under a strict 16GB VRAM budget on Nvidia Tesla P100/T4 GPUs using native 16-bit precision (FP16) and LoRA adapters ($r = 16$) (Hu et al., 2022) on Qwen2.5-VL-3B-Instruct (Bai et al., 2025). Models are trained in native FP16 rather than 4-bit quantization due to P100 hardware constraints (Appendix A.2); base weights occupy 6.0 GB and peak VRAM remains under 14.2 GB. Table 1 summarizes configurations, dataset sizes, learning rates, and runtimes.

We investigate seven alignment paradigms (full objective formulas in Appendix A): (1) **SFT** maximizes gold step log-likelihood (Eq. 3); (2) **Full-Sequence DPO** (Rafailov et al., 2023) optimizes a Bradley-Terry preference model over entire trajectories:

$$\mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) = -\mathbb{E} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right] \quad (2)$$

with $\beta = 0.1$; (3) **Suffix Step-DPO** (Lai et al., 2024) applies DPO loss solely to divergent suffixes from t^* , masking prefix tokens from gradients (Eq. 4); (4) **KTO** (Ethayarajh et al., 2024) optimizes unpaired utility over implicit rewards (Eq. 5); (5) **Two-Stage SFT→DPO** initializes policy and reference from the SFT checkpoint before DPO; (6) **SimPO** (Meng et al., 2024) replaces the reference model with length-normalized policy probabilities and a target margin γ ($\beta = 2.0$, $\gamma/\beta = 0.5$, Eq. 6), halving forward passes per step; and (7) **Pareto-DPO** trains with the identical DPO loss but filters pairs via 9-dimensional Pareto dominance across judge criteria (§2.5).

2.5 DynamicPRM

A single binary score per step cannot distinguish an axis misreading from a hallucinated legend entry. Motivated by Dynamic and Generalizable Process Reward Modeling (DG-PRM) (Yin et al., 2024), we replace the single scalar with a multi-criteria reward built from the judge’s own failure language.

Reward Tree Construction: We reuse the 9 error categories from §3.2 as parent nodes (*other/unspecified* excluded). Within each parent, we cluster failure critiques by MiniLM sentence embeddings (k -means), merge near-duplicates (cosine distance ≤ 0.14), and distill each surviving cluster into a rubric statement via one LLM call per parent. This yields **33 scored criteria** under 9 parents, requiring only 9 API calls for distillation.

Dynamic Multi-Criteria Judge: For each step, the judge inspects the 33-criterion tree and self-selects applicable criteria, scoring each on a 1–3 scale blind to ground-truth answers. Across 309 training-pool questions (1,199 rollouts, 4,652 steps), the judge selects 3.85 criteria per step on average with 0% empty selections. While collapsing these scores to a single scalar does not beat the binary judge (-2.6 pp), the dynamic judge exposes a rich 9-dimensional failure profile per rollout.

Pareto-Dominance Pair Selection: Rather than reducing scores to a scalar mean, we retain the 9-dimensional vector (one score per parent category). A preference pair (y_w, y_l) is retained only when y_w Pareto-dominates y_l : at least as good on every category and strictly better on at least one. This discards pairs trading off across disparate failure axes (e.g., fewer hallucinations but worse axis reading). Across the 309-question pool, this yields **154 pairs**, trained with the identical DPO trainer and hyperparameters as Full DPO (Eq. 2), achieving CharXiv’s top result at **30.0% exact-match** (Table 3).

Cross-Benchmark Generalization: DG-PRM claims dynamic criteria are generalizable (Yin et al., 2024). We test this directly in §3.5 by transferring the 33-criterion tree unmodified to ChartQA (Masry et al., 2022), where all 33 criteria activate at high density (4.21 criteria/step) and Pareto-DPO wins decisively.

2.6 Test-Time Verification and Search

Beyond offline dataset curation, process reward models can rerank candidate solutions at inference time (Lightman et al., 2023). Using rollouts from §2.1 without generating new traces, we com-

Strategy	Holdout EM	Δ vs. Rand	Solvable
Uniform Random	18.4%	–	41.8%
Majority Vote	21.0%	+2.6 pp	47.7%
PRM Best-of-N	27.5%	+9.1 pp	62.5%
<i>Oracle Bound</i>	<i>44.0%</i>	<i>+25.6 pp</i>	<i>100.0%</i>

Table 2: **Test-Time Verifier Performance** ($N = 309$). Selecting candidates by step process reward outperforms self-consistency by +6.5 pp.

pare uniform random selection, majority-vote self-consistency (Wang et al., 2023), and PRM best-of- N (selecting the rollout with highest mean process reward $R(\tau)$, Eq. 1) across 309 training-pool questions with multiple candidates.

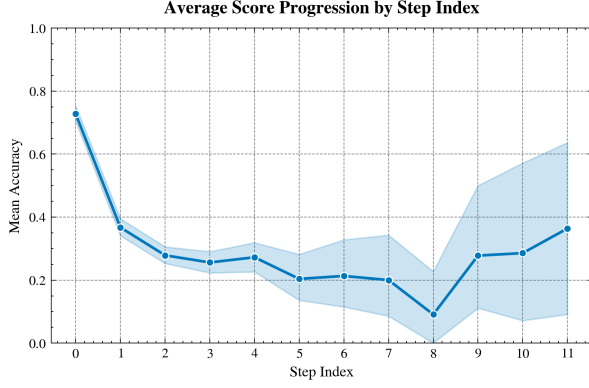
As shown in Table 2 (and Figure 10, Appendix D), PRM best-of- N reaches **27.5%** exact match, outperforming self-consistency (21.0%, +6.5 pp) and random selection (18.4%, +9.1 pp). The oracle bound (44.0%) indicates substantial headroom: while majority voting fails when the correct answer is a minority sample at $\tau = 0.7$, the PRM verifier identifies correct reasoning chains 62.5% of the time when present (27.5 / 44.0), boosting inference accuracy without additional fine-tuning.

3 Results and Analysis

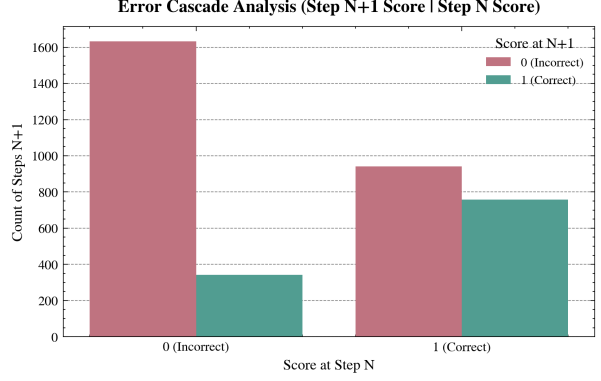
We evaluate the empirical performance of multimodal chart reasoning across four dimensions: reasoning trajectory dynamics, failure mode taxonomy, preference alignment effectiveness, and cross-dataset generalization. First, we examine multi-step rollouts from the unaligned base model, tracking the accuracy cliff and error cascades across reasoning depth (§3.1). Second, we analyze 2,920 refuted reasoning steps to establish an empirical error taxonomy (§3.2), demonstrating that failures stem overwhelmingly from visual perception rather than arithmetic. Third, we benchmark seven preference-aligned policy models on held-out scientific reasoning problems (§3.3) and analyze the Pareto trade-off between reasoning accuracy and output structure compliance (§3.4). Fourth, we retrain and re-evaluate all seven methods on a second, disjoint benchmark, ChartQA, to test whether any of these findings are specific to CharXiv (§3.5).

3.1 Reasoning Trajectory Dynamics and Error Cascades

We evaluate 1,274 rollouts (4,947 steps) generated by Qwen2.5-VL-3B-Instruct across



(a) Step Score Cliff by Reasoning Depth



(b) Error Cascade Transition Probabilities

Figure 1: **Reasoning Trajectory Dynamics.** (a) Step accuracy drops from 73% at Step 0 to 26% at Step 3. (b) When a step fails, the next fails 82.7% of the time, demonstrating catastrophic cascading from early perceptual errors.

500 CharXiv reasoning problems. Figure 1 tracks score progression and transition dynamics (full statistics in Table 6, Appendix C).

The Reasoning Cliff: Reasoning quality degrades rapidly with chain depth (Figure 1a). At Step 0 (initial inspection), the model achieves a 73.0% pass rate, falling to 43.1% at Step 1 and 26.3% by Step 3. Overall, only 12.1% of rollouts (154/1,274) complete all steps without error. Multi-step chart reasoning requires sequential grounding: isolating panels, reading axes, matching legends, and deducing trends. For a compact 3B model, maintaining visual fidelity across sequential steps proves difficult.

Catastrophic Error Cascades (82.7%): When a step fails ($s_t = 0$), the immediate next step fails **82.7% of the time** ($P(S_{N+1} = 0 \mid S_N = 0) = 82.7\%$), recovering to a valid step in only 17.3% of cases (Figure 1b). Crucially, **79.7%** of all flawed rollouts commit their first error within Steps 0–1 (31.0% at Step 0, 48.8% at Step 1).

Why Process Verification Matters: Errors do not accumulate gradually at the end of solutions; models stumble immediately. An initial misread serves as a poisoned premise: subsequent steps apply sound logic to corrupted coordinates or curves. This dynamic justifies process reward modeling (Lightman et al., 2023). Outcome models (ORMs) observe only terminal failure, whereas PRMs pinpoint the divergence step, enabling verifiers to arrest cascades before they spread.

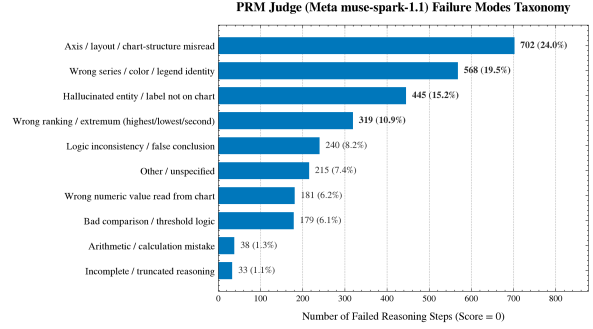


Figure 2: **Distribution of 2,920 Refuted Reasoning Steps.** Visual grounding errors (43.5%) dwarf pure arithmetic errors (1.3%).

3.2 The Perceptual Bottleneck: Why Chart Reasoning Fails

Analyzing the critiques across all 2,920 failed steps ($s_t = 0$), we categorize errors into a 9-class taxonomy (Table 5, Appendix C). Figure 2 illustrates the distribution, while Figure 6 shows temporal shifts across depth.

Perception vs. Arithmetic: A central question is whether models fail due to flawed arithmetic or faulty perception. The taxonomy provides an unequivocal answer: failures are overwhelmingly perceptual. Axis and layout misreads account for 24.0% ($N = 702$) and series/legend confusion for 19.5% ($N = 568$), totaling **43.5%** ($N = 1,270$) visual grounding errors. In sharp contrast, pure arithmetic mistakes constitute only **1.3%** ($N = 38$), or 1 in every 77 errors. The model’s computational reasoning is intact; its visual parsing is the bottleneck (e.g., confusing logarithmic for linear axes or swapping colored markers).

Temporal Shift Across Steps: As detailed in

Figure 6, error modes evolve along the trajectory. At Step 0, **78.5%** of errors are axis or layout misreads. In Steps 2–4, errors shift toward comparative and logical failures: entity hallucinations (15.2%), incorrect rankings/extrema (10.9%), and logical contradictions (8.2%). Downstream steps appear conceptually flawed, but are actually operating on corrupted visual inputs.

3.3 Alignment Benchmark on Held-Out Reasoning

We evaluate all aligned models on the 100-question CharXiv held-out test split (Table 3), measuring exact-match accuracy (EM Acc.), token match (Token), format-compliant accuracy (Str+Acc), ground-truth recall in text (Recall), structural compliance (Struct), preamble penalty (Preamb), and committed wrong answers (Wrong).

Pareto-DPO and Full DPO Lead: Pareto-filtered DPO (§2.5) attains the top performance at **30.0%** EM and **31.0%** token match (+4.0 pp over base 26.0%), while Full DPO reaches 29.0% (+3.0 pp) on the same 134-pair budget. The 1.0 pp gap is not statistically significant on $N = 100$, but both confirm that full-trajectory preference optimization provides the primary accuracy driver while preserving high structural compliance (97.0–97.2%).

The SFT Underperformance Paradox: SFT on 70 verified traces achieves only **23.0%** EM (-3.0 pp vs. base). Without negative feedback on subtle visual traps (e.g., distinguishing trend lines from bounds), SFT induces rigid template memorization rather than adaptable grounding. Reference-free SimPO matches the base model (26.0%): its training tail preference accuracy reached only 51.5%, showing that 134 pairs at one epoch provide insufficient signal without a reference model.

Step-DPO, KTO, and SFT→DPO: Suffix Step-DPO achieves **25.0%** EM, as prefix masking disrupts global output formatting. Unpaired KTO matches base EM at **26.0%** (29.0% token match) but suffers structural collapse, detailed below. Sequential SFT→DPO attains only **22.0%**, indicating that SFT warm-up introduces suboptimal inductive biases.

Validating Small-Data Alignment: Aligning a 3B VLM does not demand vast annotation corpora. Using 134 judge-curated preference pairs on a single T4 GPU, Full DPO delivers a +3.0 pp gain, and multi-criteria Pareto filtering boosts this to +4.0 pp, establishing state-of-the-art accuracy on CharXiv.

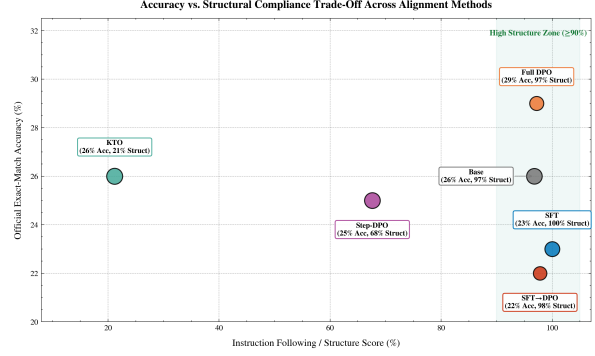


Figure 3: **Accuracy vs. Structure Pareto Frontier.** Bubble size indicates ground-truth recall in prose (47%–66%). Full DPO achieves the optimal balance (29.0% EM, 97.2% structure), while KTO attains high latent recall (66.0%) but collapses structurally (21.2%).

3.4 The Accuracy–Structure Trade-Off and Error Modes

Standard benchmarks often obscure how alignment reshapes instruction-following behavior. Figure 3 plots exact-match accuracy against structural compliance, with bubble diameters denoting ground-truth recall in text (error mode breakdowns in Figure 9b, Appendix D).

The Pareto Frontier: Figure 3 illustrates a distinct trade-off frontier. Full DPO achieves the optimal balance: 29.0% accuracy with 97.2% structural compliance and only 5.0% preamble penalty. SFT exhibits over-constrained rigidity: 100.0% format compliance with 0.0% preamble, but at the expense of reasoning flexibility (23.0% EM). Suffix Step-DPO exhibits format drift (67.6% structure, 57.0% preamble) due to prefix masking. Unpaired KTO displays a striking divergence: structural compliance collapses to 21.2% (99.0% conversational preambles), yet it achieves the highest latent ground-truth recall at **66.0%**. KTO frequently verbalizes correct values in its explanation, but fails to isolate them under `Final Answer:`.

Error Modes and Hallucination Proxy: Examining committed wrong answers (outputting a false prediction without mentioning the true value, Figure 9b), structurally disciplined models commit more decisive hallucinations: SFT commits in 43.0%, Full DPO in 49.0%, and SFT→DPO in 53.0% of cases. Trained to produce definitive answers, they cannot express visual uncertainty. In contrast, KTO exhibits the lowest committed wrong rate (30.0%), as its conversational style explores alternatives.

Model	Supervision	EM Acc.	Token	Str+Acc	Recall	Struct	Preamb	Wrong
Qwen2.5-VL-3B	Base Instruct	26.0%	28.0%	26.0%	63.0%	96.8%	7.0%	37.0%
SFT	Gold Traces	23.0%	28.0%	28.0%	57.0%	100.0%	0.0%	43.0%
Full DPO	Chosen vs. Rej.	29.0%	30.0%	29.0%	51.0%	97.2%	5.0%	49.0%
Step-DPO	Prefix-Masked	25.0%	25.0%	16.0%	64.0%	67.6%	57.0%	36.0%
KTO	Unpaired	26.0%	29.0%	0.0%	66.0%	21.2%	99.0%	30.0%
SFT→DPO	Two-Stage	22.0%	25.0%	25.0%	47.0%	97.8%	0.0%	53.0%
<i>Specialist & Reference-Free Extensions:</i>								
ChartGemma (3B)	Specialist	27.0%	27.0%	—	—	—	—	—
SimPO	Ref.-Free	26.0%	27.0%	26.0%	—	95.0%	6.0%	40.0%
Pareto-DPO	Multi-Filter	30.0%	31.0%	30.0%	—	97.0%	4.0%	46.0%

Table 3: **Main Holdout Benchmark Evaluation on CharXiv** ($N = 100$). EM Acc. is official exact-match accuracy; Token is normalized token match; Str+Acc is format-compliant and correct; Recall measures whether the ground truth is mentioned in text; Struct is structural compliance (0–100%); Preamb is preamble penalty rate; Wrong is wrong committed answers (hallucination proxy).

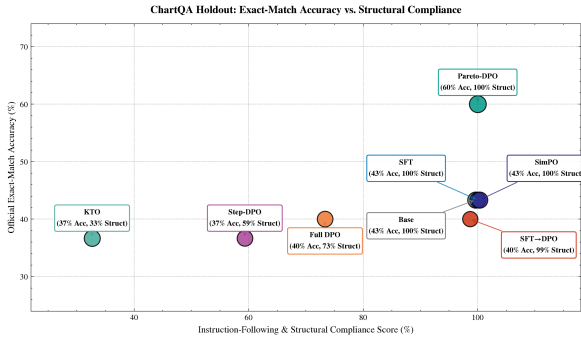


Figure 4: **ChartQA Accuracy vs. Structure Trade-Off.** While CharXiv exhibits a dispersed frontier, on ChartQA Pareto-DPO dominates outright (60.0% EM, 100% structure).

3.5 Cross-Dataset Generalization: ChartQA Holdout

To determine whether our findings reflect general properties rather than CharXiv-specific artifacts, we evaluate the entire pipeline on a second, disjoint benchmark: **ChartQA** (Masry et al., 2022). We transfer the 33-criterion reward tree of §2.5 unmodified—no retuning, no new criteria, and zero additional judge distillation calls. All scoring runs on gemini-3.5-flash-lite, ensuring fair cross-dataset comparison.

We sample 150 human-authored reasoning questions from ChartQA’s test set for training and hold out 30 for evaluation, matching CharXiv’s focus on non-templated reasoning. All seven alignment methods are retrained from scratch under identical hyperparameters (Table 1) and evaluated across the same metrics (Table 4).

Pareto-DPO Outperforms Across All Metrics:

As shown in Table 4 (and Figure 11, Appendix E), Pareto-DPO achieves **60.0%** exact match, surpass-

ing base by +16.7 pp—the largest margin across both benchmarks, though at $N = 30$ this margin is not statistically robust (see [Limitations](#)). It simultaneously attains peak token match (73.3%), 100% structure compliance, and 73.3% latent recall. Unlike CharXiv where Full DPO trailed narrowly, on ChartQA Pareto-DPO dominates completely (Figure 4), while Full DPO reaches only 40.0%.

Consistency of Alignment Behaviors: SFT (43.3%) and SimPO (43.3%) match base performance, mirroring CharXiv. SimPO’s tail preference accuracy (51.6%) closely replicates its CharXiv level (51.5%), underscoring objective limits under low pair budgets. Step-DPO (36.7% EM, 59.3% structure) and KTO (36.7% EM, 32.7% structure, 86.7% preamble) reproduce their formatting collapse. KTO again yields the lowest committed wrong rate (23.3%).

Why Pareto-DPO Transfers: Training loss convergence does not guarantee holdout reasoning: SFT→DPO achieved 76.6% training preference accuracy on ChartQA but reached only 40.0% holdout EM, trailing Pareto-DPO (60.9% training accuracy). Pareto filtering discards rollouts with cross-category trade-offs, providing unambiguous contrastive signal. Crucially, the 33-criterion reward tree transfers seamlessly without retuning (firing at 4.21 criteria/step), proving that dynamic PRM criteria generalize across visual reasoning distributions.

4 Conclusions

We investigated process reward modeling and preference optimization for complex chart reasoning on compact 3B vision-language models under consumer GPU constraints. Diagnostic anal-

Model	Supervision	EM Acc.	Token	Str+Acc	Recall	Struct	Preamb	Wrong
Qwen2.5-VL-3B	Base Instruct	43.3%	60.0%	60.0%	66.7%	100.0%	0.0%	33.3%
SFT	Gold Traces	43.3%	56.7%	56.7%	63.3%	100.0%	0.0%	36.7%
Full DPO	Chosen vs. Rej.	40.0%	50.0%	26.7%	63.3%	73.3%	46.7%	33.3%
Step-DPO	Prefix-Masked	36.7%	46.7%	20.0%	63.3%	59.3%	56.7%	33.3%
KTO	Unpaired	36.7%	50.0%	0.0%	66.7%	32.7%	86.7%	23.3%
SFT→DPO	Two-Stage	40.0%	53.3%	53.3%	60.0%	98.7%	0.0%	40.0%
<i>Reference-Free and Multi-Filter Extensions:</i>								
SimPO	Ref.-Free	43.3%	60.0%	60.0%	66.7%	100.0%	0.0%	33.3%
Pareto-DPO	Multi-Filter	60.0%	73.3%	73.3%	73.3%	100.0%	0.0%	26.7%

Table 4: **Cross-Dataset Holdout Evaluation on ChartQA** ($N = 30$). Evaluated on human-authored reasoning problems. Pareto-DPO achieves top scores across all metrics except Wrong rate.

ysis of 2,920 refuted steps revealed an empirical perceptual bottleneck: 43.5% of errors originate from visual grounding (axes, legend styles) rather than arithmetic (1.3%), cascading with an 82.7% failure continuation rate. On both CharXiv and ChartQA, Pareto-dominance-filtered DPO consistently achieves the strongest reasoning accuracy among the seven alignment methods evaluated (30.0% and 60.0% EM) using fewer than 155 contrastive pairs. Furthermore, the 33-criterion dynamic reward tree transfers across benchmarks without modification, validating the generalizability of structured process supervision.

Limitations

Our study evaluates a 3B parameter model (Qwen2.5-VL-3B); scaling dynamics on 7B–72B architectures remain to be explored. Second, while CharXiv holdout evaluations use $N = 100$ problems, ChartQA transfer is evaluated on only $N = 30$ questions due to API and compute constraints: at this scale, a single question flip moves exact-match by 3.3 pp, and a 5-question flip, plausible sampling noise, would move it by 16.7 pp—most of Pareto-DPO’s entire margin over the base model. The ChartQA result should be read as suggestive of transfer, not as a statistically established margin. Finally, rollout generation relies on teacher-free sampling; integrating Monte Carlo Tree Search at inference time presents an exciting avenue for future work.

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A Alignment Objectives, Optimization Details, and Training Dynamics

In this section, we provide the full mathematical formulations, training hyperparameters, and empirical loss trajectories for the preference alignment paradigms investigated in §2.4.

A.1 Mathematical Formulations

1. Supervised Fine-Tuning (SFT): Demonstrations are trained by maximizing the conditional log-likelihood of verified gold reasoning steps:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x, y_w)} \sum_{t=1}^{|y_w|} \log \pi_{\theta}(y_{w,t} \mid x, y_{w,<t}) \quad (3)$$

2. Full-Sequence DPO: Sequence-level Direct Preference Optimization (Rafailov et al., 2023) is defined in the main text (Equation 2) with Bradley-Terry parameter $\beta = 0.1$.

3. Suffix Step-DPO: Suffix Step-DPO (Lai et al., 2024) isolates the credit assignment problem by masking the shared prefix $y_{<t^*}$ out of the gradient, applying the DPO loss strictly to the divergent suffix starting at the first point of divergence t^* :

$$\mathcal{L}_{\text{Step-DPO}}(\theta) = -\mathbb{E} \left[\log \sigma \left(\beta \sum_{k \geq t^*} \log \frac{\pi_{\theta}(y_{w,k})}{\pi_{\text{ref}}(y_{w,k})} - \beta \sum_{k \geq t^*} \log \frac{\pi_{\theta}(y_{l,k})}{\pi_{\text{ref}}(y_{l,k})} \right) \right] \quad (4)$$

4. Kahneman-Tversky Optimization (KTO): KTO (Ethayarajh et al., 2024) optimizes policy weights on unpaired desirable ($y \in Y_{\text{pos}}$) and undesirable ($y \in Y_{\text{neg}}$) reasoning traces via prospect-theoretic value functions:

$$\mathcal{L}_{\text{KTO}}(\theta) = \mathbb{E}_y [w(y) (1 - v_{\beta}(\hat{r}_{\theta}(x, y) - z_{\text{ref}}))] \quad (5)$$

where $\hat{r}_{\theta}(x, y) = \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)}$ is the implicit reward, $v_{\beta}(r) = \sigma(\beta r)$ for desirable rollouts, $v_{\beta}(r) = \sigma(-\beta r)$ for undesirable rollouts, $w(y)$ is an importance weight ensuring balanced gradients, and z_{ref} is a running exponential moving average estimating the KL divergence baseline.

5. Simple Preference Optimization (SimPO): SimPO (Meng et al., 2024) eliminates the reference policy forward pass by substituting reference log-probabilities with length-normalized policy log-

probabilities and an explicit target margin γ :

$$\mathcal{L}_{\text{SimPO}}(\theta) = -\mathbb{E} \left[\log \sigma \left(\frac{\beta}{|y_w|} \log \pi_{\theta}(y_w \mid x) - \frac{\beta}{|y_l|} \log \pi_{\theta}(y_l \mid x) - \gamma \right) \right] \quad (6)$$

We set $\beta = 2.0$ and target margin $\gamma = 1.0$ ($\gamma/\beta = 0.5$).

A.2 Optimization Details and Hardware Footprint

All fine-tuning experiments were conducted under a strict 16GB VRAM hardware ceiling on an Nvidia Tesla P100 GPU (Kaggle) or Nvidia Tesla T4 GPU (Google Colab). To operate reliably within this compute constraint:

- **Native FP16 Precision:** Models are trained in 16-bit half precision (`torch.float16, use_4bit=False`). Our Kaggle Nvidia Tesla P100 GPUs (Pascal architecture, compute capability 6.0) lack hardware support for `bitsandbytes` 4-bit NF4 CUDA kernels (requiring compute capability ≥ 7.0). Base weights occupy ≈ 6.0 GB, training with zero quantization distortion.
- **LoRA Adaptation:** Applied to all linear projections with rank $r = 16$, $\alpha = 32$, and dropout 0.05 (Hu et al., 2022).
- **Memory Optimizations:** Gradient checkpointing and micro-batch size 1 keep peak memory < 14.2 GB throughout training.

A.3 Empirical Training Dynamics

Figure 5 visualizes the optimization dynamics across paradigms. In SFT (top-left), cross-entropy loss decays smoothly from 1.8 to 0.4 within one epoch, verifying prompt structure memorization. In Full DPO (top-right), loss decreases steadily with stable margin separation. In contrast, Step-DPO (bottom-left) exhibits rapid implicit margin growth ($\Delta r \rightarrow +10.0$), because masking the shared prefix concentrates probability updates onto fewer tokens. Finally, KTO (bottom-right) displays margin oscillations due to single-sample reward variance in unpaired estimation, which contributes to its subsequent format collapse.

B Prompt Protocol and Output Parsing

B.1 Structured Chain-of-Thought Prompt

To facilitate reliable process reward scoring, we design a structured Chain-of-Thought template that

Training Dynamics & Loss Trajectories Across Fine-Tuning Paradigms

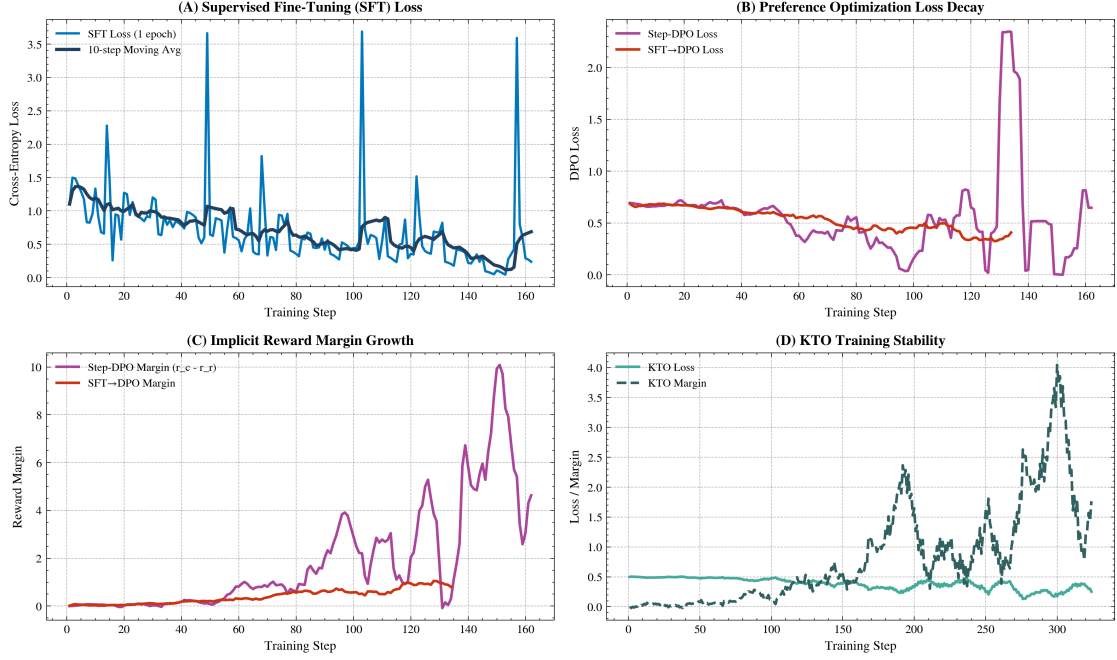


Figure 5: **Training Dynamics Across Alignment Paradigms.** Top-left: SFT cross-entropy loss decay ($1.8 \rightarrow 0.4$). Top-right: Sequence DPO loss trajectories converging smoothly. Bottom-left: Step-DPO implicit reward margin explosion ($\Delta r \rightarrow +10.0$) caused by prefix masking. Bottom-right: KTO margin oscillations under single-GPU micro-batch constraints.

enforces explicit step boundaries and separates visual inspection from numerical deduction:

```
Solve the chart reasoning
question step by step.
Structure your answer strictly
as follows:
Step 1: <visual observation>
Step 2: <intermediate reading
or deduction>
...
Final Answer: <concise answer>
```

We enforce three practical formatting requirements during rollout evaluation:

1. **Indexed Steps:** Every reasoning step must begin with an explicit numeric label (Step 1:, Step 2:, etc.) followed by a newline.
2. **Isolated Final Answer:** The final answer must appear on its own terminal line prefixed with Final Answer:, containing solely the extracted value or category without markdown asterisks or surrounding quotation marks.
3. **Preamble Suppression:** Generations must omit conversational filler (e.g., “Certainly! Let me analyze this plot.”). Any non-compliant opening text is recorded as a preamble violation.

B.2 Automated Answer Parsing and Normalization

Evaluation rollouts are parsed by searching for the regex pattern `r"Final Answer:\s*(.+)"` from the end of the text. Extracted answer strings undergo standard lowercase whitespace normalization (removing trailing punctuation, dollar signs, percent signs, and converting multi-space sequences to single spaces). Exact-match accuracy evaluates whether the normalized prediction matches the ground-truth string.

C Diagnostic PRM Judge, Error Taxonomy, and Cascades

C.1 Multimodal Rollout-Batched Judge

Evaluating 4,947 reasoning steps individually via multimodal API endpoints presents severe latency and cost bottlenecks. We implement *rollout-level batching*: chart images are resized to a maximum dimension of 512 pixels (preserving axis tick marks and data markers) and bundled with the complete sequence of generated steps into a single API invocation to `muse-spark-1.1` (Meta Superintelligence Labs, 2026; Zheng et al., 2023). The judge outputs a structured JSON list containing step-level

Error Category	Count	Share	Diagnostic Keywords	Representative Judge Critique Excerpt
Axis / Layout Misread	$N = 702$	24.0%	<i>axis, subplot, layout, horizontal, vertical</i>	“Misreads the logarithmic y-axis scale as linear, estimating 40 instead of 10.”
Wrong Series / Color	$N = 568$	19.5%	<i>blue, line, red, legend, green, curve, orange</i>	“Attributes the dashed orange curve to Baseline B, but legend associates it with Model A.”
Hallucinated Entity	$N = 445$	15.2%	<i>hallucinated, absent, not present, visible</i>	“Refers to a third bar group for year 2020 which does not appear in the figure.”
Wrong Ranking / Extremum	$N = 319$	10.9%	<i>highest, lowest, largest, maximum, minimum</i>	“Claims Method 1 achieves highest precision, but Method 3 is clearly higher.”
Logic Inconsistency	$N = 240$	8.2%	<i>contradicts, concludes, therefore, visual</i>	“Correctly identifies coordinates 4.2 and 5.1, but falsely concludes $4.2 > 5.1$.”
Wrong Numeric Value	$N = 181$	6.2%	<i>misreads, chart shows, near, false reading</i>	“Reads point height as 0.65 when the marker aligns with the 0.50 gridline.”
Bad Comparison / Logic	$N = 179$	6.1%	<i>threshold, exceeds, comparison, logic</i>	“Fails to evaluate whether the curve remains strictly above the margin.”
Arithmetic Mistake	$N = 38$	1.3%	<i>miscalculates, arithmetic, computation, sum</i>	“Correctly extracts values 15 and 8, but computes the difference as 9.”
Incomplete / Truncated	$N = 33$	1.1%	<i>truncated, incomplete, cuts off, empty</i>	“Generation terminates prematurely mid-sentence without a verifiable step.”
Other / Unspecified	$N = 215$	7.4%	<i>unspecified, error, incorrect</i>	“Vague failure critique without explicit grounding attribution.”
Perceptual Total	$N = 1,270$	43.5%	<i>Combined Axis Misread (24.0%) and Series/Legend Identity (19.5%)</i>	
Arithmetic Total	$N = 38$	1.3%	<i>Pure calculation failure accounts for only 1 in 77 step errors</i>	

Table 5: **PRM Judge Error Taxonomy from 2,920 Refuted Reasoning Steps.** Categorized via priority-regex analysis over natural language critiques from muse-spark-1.1. Visual grounding (axes, series colors) forms the primary reasoning bottleneck, while arithmetic mistakes are rare.

binary scores $s_t \in \{0, 1\}$ and qualitative critique sentences for all refuted steps.

C.2 Error Taxonomy Analysis

Table 5 catalogues the 9-class empirical error taxonomy extracted from 2,920 failed steps. Figure 6 visualizes how error composition shifts over reasoning depth. In Step 0, axis and layout misreads dominate at 78.5%, reflecting the initial perceptual difficulty of parsing logarithmic axes, dual scales, and multi-panel arrangements. As reasoning chains progress into Steps 2–4, errors transition into entity hallucinations (15.2%) and logical contradictions (8.2%), which stem directly from the model attempting to build deductions upon corrupted initial observations.

C.3 Trajectory Cascades and Domain Distributions

Table 6 summarizes global trajectory statistics across the 500 CharXiv reasoning pool. Over 79.7% of flawed rollouts make their first mistake in Steps 0 or 1. Once a step fails, the subsequent step fails with 82.7% probability ($P(S_{N+1} = 0 \mid S_N = 0) = 82.7\%$). Figure 7 maps the co-occurrence of error categories: series identity confusion frequently co-occurs with downstream logical inconsistency (460 instances).

Trajectory Metric	Count	Share
CharXiv Reasoning Problems	500	100.0%
Evaluated Rollout Traces	1,274	100.0%
Mean Candidates / Question	3.83	–
Total Judged Reasoning Steps	4,947	100.0%
Step Verification:		
– Step Pass Rate (Score = 1)	2,027	41.0%
– Step Failure Rate (Score = 0)	2,920	59.0%
Rollout Outcome:		
– Fully Valid Rollouts (All steps = 1)	154	12.1%
– Rollouts with ≥ 1 Step Error	1,120	87.9%
Mean Steps per Rollout	3.88	–
Failure Progression Dynamics:		
– First Error at Step 0	347	31.0%
– First Error at Step 1	546	48.8%
– First Error at Step ≤ 1 Cumulative	893	79.7%
– Cascade: $P(S_{N+1} = 0 \mid S_N = 0)$	1,630	82.7%
– Recovery: $P(S_{N+1} = 1 \mid S_N = 0)$	340	17.3%

Table 6: **Reasoning Trajectory Statistics on CharXiv 500 Pool.** Over 79% of trajectory errors begin within the first two steps. Once an error occurs, downstream steps fail 82.7% of the time.

Figure 8 breaks down error rates across academic disciplines: Physics and Economics exhibit the highest rate of axis misreads ($>34\%$), reflecting complex scientific tick scales, while Computer Science and Mathematics exhibit elevated entity hallucinations ($>22\%$).

Evolution of PRM Judge Error Classifications Across Reasoning Steps

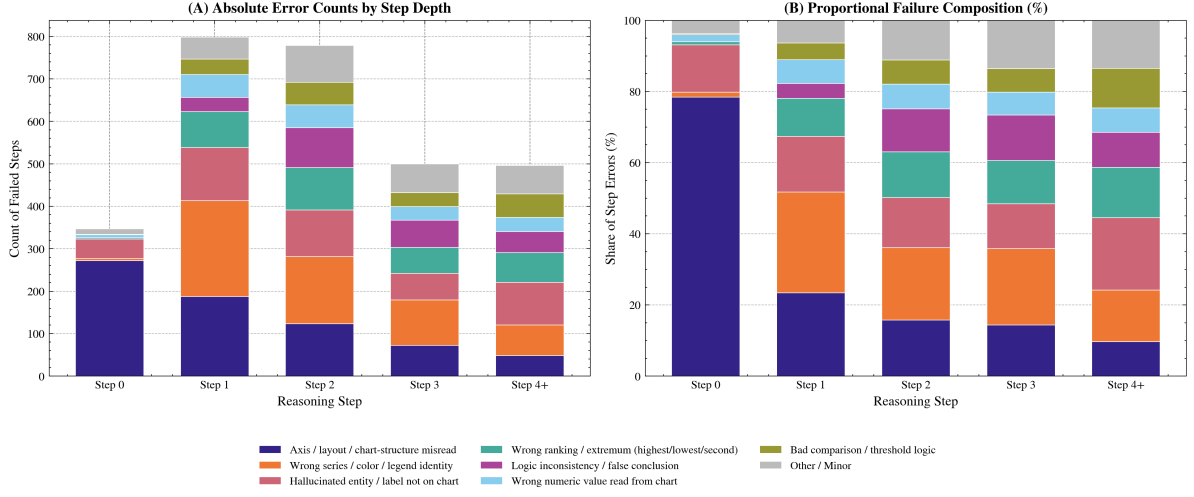


Figure 6: **Error Mode Composition Across Step Depth.** Detailed breakdown of 2,920 refuted reasoning steps evaluated by muse-spark-1.1 across reasoning steps 0 to 4+. Step 0 is dominated by axis and layout misreads (78.5%), which triggers cascading entity hallucinations and logical inconsistencies in downstream steps.

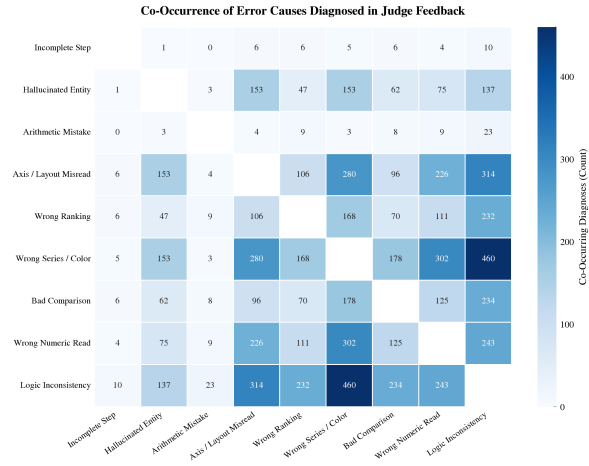


Figure 7: **Compound Error Co-Occurrence Heatmap.** Off-diagonal counts show strong coupling between perceptual errors and downstream logical failures (460 co-occurrences between logic errors and series confusion).

D Structural Instruction Compliance, Error Modes, and Test-Time Search

D.1 Instruction Following across Tags

Figure 9a breaks down structural compliance across individual formatting tags (Step 1:, intermediate steps, Final Answer:, and preamble absence). SFT achieves 100% compliance across all tags through prompt memorization. Full DPO maintains 97.2% overall structural compliance with only 5.0% conversational preambles. Conversely, KTO collapses to 21.2% compliance: 99.0% of KTO responses contain conversational introduc-

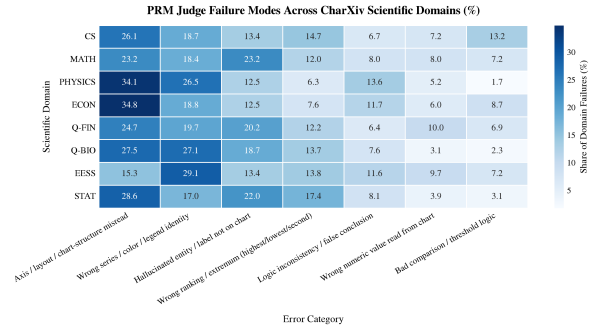
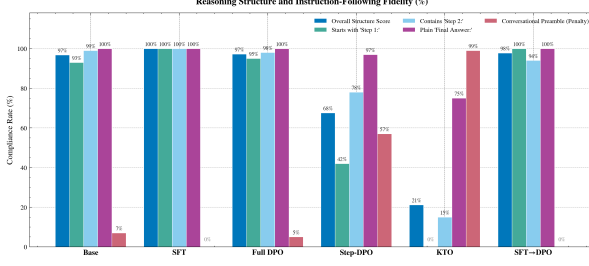


Figure 8: **Error Distribution Across CharXiv Domains.** Physics and Economics charts exhibit the highest rate of axis misreads (>34%), while Mathematics and Computer Science suffer more from entity hallucinations (>22%).

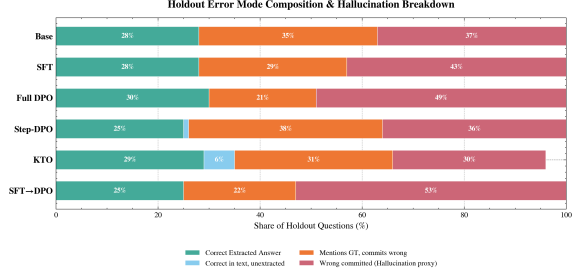
tory phrases and 78.8% fail to include the required final answer delimiter.

D.2 Error Mode Breakdown and Hallucination Proxy

Figure 9b partitions test rollouts into four mutually exclusive categories: extracted correct, unextracted correct (the true value appears in reasoning but is omitted from the final answer tag), partial mention (overlapping substrings), and committed wrong answers (the model commits to an incorrect number without mentioning the ground truth). Models with high structural discipline exhibit higher committed hallucination rates (SFT 43.0%, Full DPO 49.0%, SFT→DPO 53.0%), while KTO exhibits the lowest committed rate (30.0%) by hedging in prose.



(a) Instruction Following Compliance by Tag



(b) Holdout Error Modes & Hallucination Proxy

Figure 9: Instruction Following and Error Mode Breakdown. (a) Structural compliance across individual prompt constraints (Step 1 :, intermediate steps, Final Answer :, and preamble suppression). SFT achieves 100% across all tags, while KTO collapses to 21.2% compliance primarily due to conversational preambles (99.0%) and missing answer delimiters (78.8%). (b) Partitioning test responses into extracted correct, unextracted correct in prose, partial ground-truth mentions, and committed wrong answers (hallucination proxy).

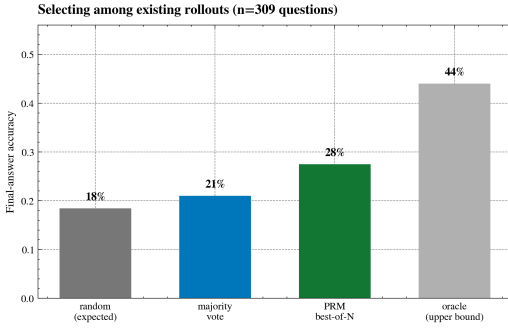


Figure 10: Inference-Time PRM Verifier Accuracy. Average step process reward selection (27.5%) outperforms self-consistency majority voting (21.0%) and random rollouts (18.4%) across 309 multi-candidate questions.

D.3 Inference-Time Candidate Search

Figure 10 illustrates the accuracy of inference-time candidate selection across 309 questions with multiple candidate rollouts. Selecting the candidate with highest mean step process reward $R(\tau)$ achieves 27.5% exact match, outperforming majority-vote self-consistency (21.0%, +6.5 pp) and uniform random selection (18.4%, +9.1 pp), confirming that offline process reward models serve as effective zero-shot inference rerankers.

E Cross-Dataset Generalization to ChartQA

E.1 Transfer Methodology and Dataset Selection

To test whether the 33-criterion dynamic reward tree generalises beyond CharXiv, we evaluate zero-retuning transfer on ChartQA (Masry et al., 2022). We isolate 150 human-authored reasoning questions from ChartQA’s test split for fine-tuning and

hold out 30 questions for evaluation. Human-authored questions test complex multi-step reasoning over real-world charts, unlike machine-generated questions that use synthetic templates.

E.2 Tree Activation and Generalization Results

The 33-criterion tree transfers unmodified to ChartQA without altering rubrics or parent categories. Across 612 rollouts on ChartQA, the dynamic judge (gemini-3.5-flash-lite) selects an average of 4.21 criteria per step with 0% empty selections, confirming rubric generalizability.

Figure 11 presents the holdout comparison. Pareto-DPO achieves 60.0% exact match, 73.3% token match, and 100.0% structural compliance, outperforming the base model (43.3% EM) by +16.7 pp. In contrast, Full DPO reaches 40.0% EM (73.3% structure) and Step-DPO reaches 36.7% EM (59.3% structure). Pareto filtering eliminates rollouts with cross-criterion trade-offs, yielding a clean contrastive signal that transfers robustly.

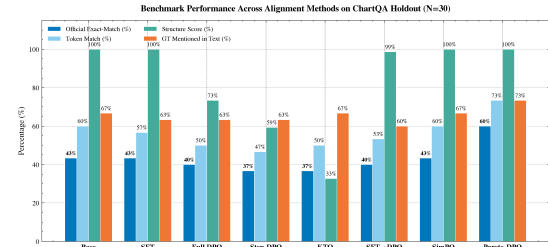


Figure 11: ChartQA Holdout Benchmark ($N = 30$). Pareto-DPO leads on exact-match, token match, and structure score simultaneously, matching the pattern reported numerically in Table 4.